

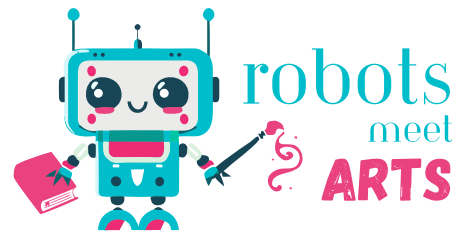


Co-funded by
the European Union

Project number: 2023-1-
FR01-KA220-SCH-
000151881

DANCE MODE!





Subject:

Sport

Content:

Develop your motor skills and learn to express yourself using your body. Learn methods and tools through physical practice.

Computational practices: choose what fits

- Decomposition
- Pattern recognition
- Abstraction
- Algorithms

Unplugged/plugged:

Unplugged



Preparation time:

10 min



Age:

7-12 years



Duration:

30 min



Location:

Outside/Inside



Group distribution:

Groupwork in group of 4-5



Difficulty



After this lesson, the children will be able to...

"Develop your motor skills and learn to express yourself using your body"

The activity translates computer language into precise body movements, allowing students to develop their coordination, spatial orientation and bodily expressiveness while mastering movement sequences of increasing complexity.

"Appreciate methods and tools through physical practice"

Students physically embody abstract computer concepts (sequences, loops, conditions), making these notions tangible and memorable through bodily experience, while learning to formalize and analyze movement methodically.

"Share rules, assume roles and responsibilities"

Through teamwork, the exchange of choreographies, and collective assessment, students develop their ability to respect common conventions, assume specific responsibilities, and collaborate effectively to achieve a shared goal.

"Practicing languages"

The activity intuitively introduces the fundamentals of computational thinking. Students discover the logic of Scratch blocks (sequences, loops, conditions) through tangible experience before encountering them on screen, facilitating the later transition to digital programming.

Materials needed:

- Introductory game “Memory Scratch”
- Scratch block card games (one game per group)
- Visual support to explain Scratch blocks (posters or presentations)
- Stopwatch to manage time
- Paper, scissors, markers for the creative phase
- Possibly: tablet or camera to document the choreographies

Helpful links:

- For more in-depth practice of unplugged programming and block programming:
 - Le Erasmus + project **Unplugged** : <https://www.unplugged-quest.eu/>
 - **Scratch Jr** (for young children ages 5-7): <https://www.scratchjr.org/>
- **Scratch** (for children ages 8-18): <https://scratch.mit.edu/>
 - **Tutorials for children**: <https://www.youtube.com/watch?v=svC8TGqPHhE>, <https://www.youtube.com/watch?v=tJNzdLChCe8>
 - **Code.org** : <https://code.org/>
 - **Kodable** : <https://www.kodable.com/>
- To explore the activity on the interdisciplinary side:
 - Learn by moving and dancing: www.gonoodle.com.
 - More ideas on movement games: <https://earlyimpactlearning.com/mindful-movement-for-kids-14-games-tips/>
 - Learn about kinesthetic intelligence: <https://study.com/learn/lesson/kinesthetic-intelligence-skills.html>

Preparation and teacher notes

Prepare pairs of cards for the "Memory" game: one card with **the image of the Scratch** block and another with its **description/function**. The cards are available in the appendix in the "Printable Resources" section. Plan to have several sets of cards so you can play in small groups.

Warm up

Introduce the sequence

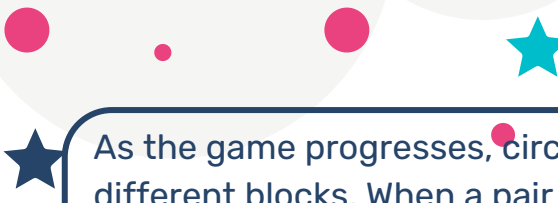
Begin the session with an **introduction to the concept of block programming**, which you can apply to Scratch, which will be used to illustrate the process in the following sequences. This activity can be applied to all block programming software suitable for cycle 3: MakeCode, MakeCode Arcade, Scratch, Kodable, Tynker, etc.

Explain to students that programming is about **giving a computer precise instructions to perform specific actions**. Introduce Scratch as an educational tool that allows students to **create stories, games, and animations by assembling colored blocks, like puzzle pieces**.

Scratch Memory Game

To further explore these concepts, play a game of Memory Scratch to help students familiarize themselves with the vocabulary and appearance of the blocks before translating them into movements. Explain that the goal is to match each Scratch block image with the word that describes it. The game consists of **32 cards, meaning 16 associations**. You can vary the level of difficulty by varying the number of cards used.

Organize the class into groups and explain the rules of Memory Scratch. Students take turns turning over two cards. If they find a correct pair, they keep it and play again. The player with the most pairs at the end wins. You can also choose a collaborative game mode, the goal being to introduce them to the blocks in a fun way.



As the game progresses, circulate between groups to help students understand the different blocks. When a pair is discovered, take the opportunity to briefly explain the block's function: "This '**Repeat 10 times**' block allows you to perform an action multiple times without having to rewrite the instruction."

Reflection on the warm up sequence

After the Memory game, gather all the students and open the Scratch editor (<https://scratch.mit.edu/projects/editor/>) to explore it collectively and make the game concrete. You can also use posters such as:

<https://edutechwiki.unige.ch/fmediawiki/images/3/37/Blocs-de-Scratch.png>.

Engage in a dynamic discussion for each block to explore its function through logic. **Add to students' answers if necessary. Ask them how certain blocks could be translated into body movement in future sessions.**

You can then open a broader discussion to stimulate their thinking:

- What did you learn about Scratch blocks today? Which ones seem easiest to understand? Which ones seem more difficult?
- What are repeat blocks like 'Repeat 10 times' or 'Repeat forever' used for? Why are they useful in programming?
- How could you use these blocks to create an interesting sequence of movements?

Finally, establish the link with the next session by announcing that some of these blocks will be transformed into body movements to concretely understand how a computer program works.



Preparation and teacher notes

Prepare several sets of cards representing the Scratch blocks that will be used for the choreography (at least 2 sets, 1 for each team). You should focus on the **blocks that can be translated into movements**:

When the green flag is clicked, Repeat indefinitely, Wait ... seconds, Repeat ... times, Turn right ... degrees, Turn left ... degrees, Move forward ... steps, Go to random position, Stop all sounds, Add ... to volume, Play meow sound, Play meow sound until end.

These cards can be printed and cut out for ease of use and are available in the "Printable Resources" section.

If possible, do the activity outside or in a gym to have the space you need to move and express yourself.

You can print the poster reminding the correspondence between each block and the associated movement so that students can easily understand and use the cards during activities.

Build-up

Begin the session in the yard or gym with a reminder of the objective: to create Scratch programs and execute them with the body, taking advantage of the available space.

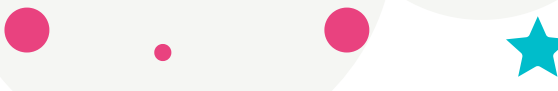
Divide the class into two groups and distribute an identical set of Scratch cards to each group. Explain that the two groups will work in parallel and exchange their creations, taking turns being the **programmer** and the **executor**.

Each group uses the cards they've been given to **program a choreography**. Let them discover the principles of constructing a logical sequence on their own, without providing too much guidance at first. This discovery approach will allow them to understand the importance of a coherent structure through experience.

Both groups work simultaneously on creating their programs for **about 15 minutes**. Circulate between the groups to observe their progress, but only intervene if there is a significant blockage.

Once the groups have prepared their sequence, they exchange them and move into **DANCE MODE**. The groups have a few minutes to discuss their approach based on the block sequence they received and rehearse.

Determine (for example by drawing lots, or on a voluntary basis) who will start dancing.



The first group then presents their choreography to the class. Students must **decode and understand the sequence of blocks**, then **apply it by moving their bodies**. Take advantage of the outdoor space to encourage large and expressive movements. The teacher can intervene if they make a mistake, similar to **live debugging**, to help them correct their interpretation and continue the choreography or stop it if a block is missing (for example, the starting block).

After the performance of the first group, the second group in turn presents the choreography it has prepared from the sequence of the other group using the same methods.

After each group's performance, the students and teacher assess the **accuracy of the dance** in relation to the given sequence. Collectively discuss the strengths and challenges encountered in performing the different blocks.

You can organize several rounds of programming and choreographic execution to allow students to **integrate the lessons from previous trials**. During iterations, you can ask specific questions: Have you considered including a starting block? How do you know when to begin the choreography without one? ... This collective reflection will allow students to discover for themselves the importance of a clear and complete structure.

Reflection on the build-up sequence

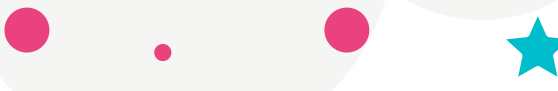
End the session with a group **discussion about your experience**. You can remain outside in a circle on the floor to create a relaxed atmosphere conducive to sharing impressions.

Guide the reflection on the **links between programming and movement**, by asking the students about their feelings in the two roles: program creators and executors.

What challenges did they encounter in each role? How does this experience help them understand the relationship between a programmer and a computer?

Explore with them the problems they encountered in the first round and how they solved them in subsequent rounds. This process of iteration and improvement is at the heart of the programming process. **What lessons did they learn from their initial mistakes? What principles did they discover for themselves?**





Preparation and teacher notes

For this last sequence, prepare blank cards for creating personalized blocks (on plain or laminated paper), as well as drawing materials (felt-tip pens, pencils, rulers).

Organize the space to allow for creative work and choreography presentations.

Also ask them how the space influenced the way they planned and executed the movements. Did they create different choreographies than they would have imagined in class? Was the range of motion greater?

Emphasize the importance of **precision in programming** and **highlight the usefulness of repetition structures**, showing how loops simplify and structure the program.

Rehearsal

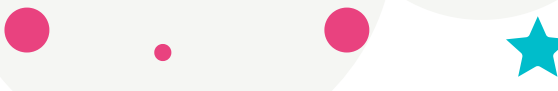
Begin the session by reviewing the previous sequences, then introduce this new creative phase in which students will **invent their own Scratch blocks and integrate them into original choreographies**. This sequence will **stimulate students' imagination and creativity**.

Keep the class divided into two groups as in the previous sequence. Distribute blank cards to each group and explain the concept of **creating custom blocks in programming, which is comparable to inventing new words in a language**. Each group will have to create 4 to 6 new, original blocks, each with a symbol, a clear name, and a precise definition of the movement represented.

Allow students to create freely. If you feel the need, you can stimulate discussion by suggesting ideas such as "Spinning like a top" or "Making a wave."

After about 15 minutes of creating, have each group present their blocks to the other group, **showing the card, explaining the name, and demonstrating the movement**. Have the whole class practice each new movement to ensure understanding.





Next, have each group create a choreography that incorporates both **the standard Scratch blocks and the new blocks created by both groups**. As in the previous step, once the sequences are created, the groups exchange them, discover them, and prepare their choreography for a few minutes, then present their sequences to the class. **Perform several rounds of the choreography** again to **stimulate the acquisition of knowledge and methods**. After each performance, discuss collectively the **accuracy of the execution compared to the initial programming**, particularly noting the **interpretation of the new personalized blocks**.

Reflection on the Rehearsal sequence

To conclude the “Dance Mode” activity, bring students together for an in-depth discussion on the connections between physical movement and program execution.

Encourage students to consider **how this hands-on approach transforms their understanding of programming**. How did creating their own blocks help them understand the concept of **modularity** and **code reuse**?

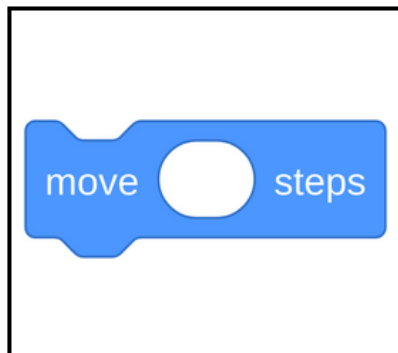
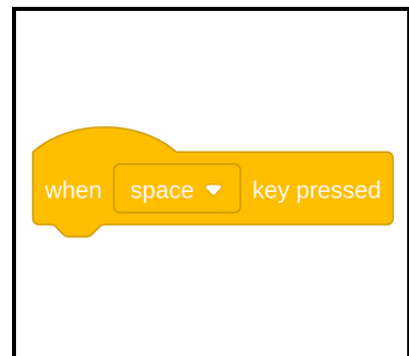
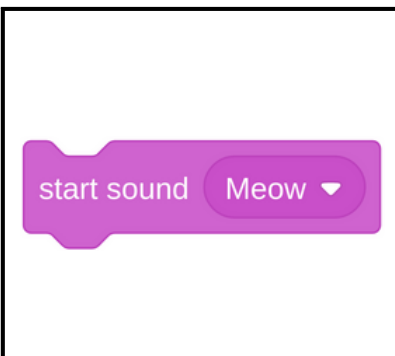
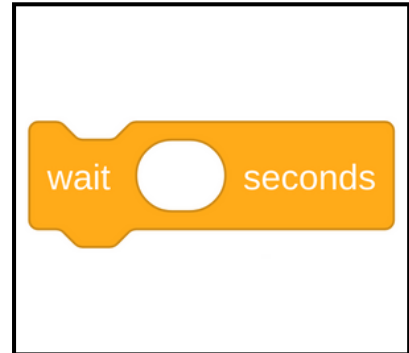
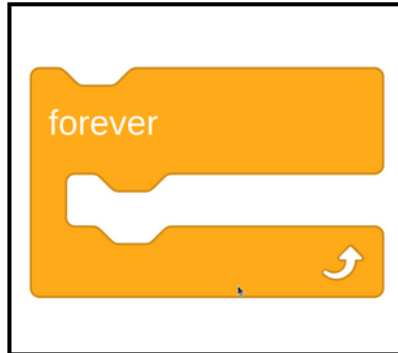
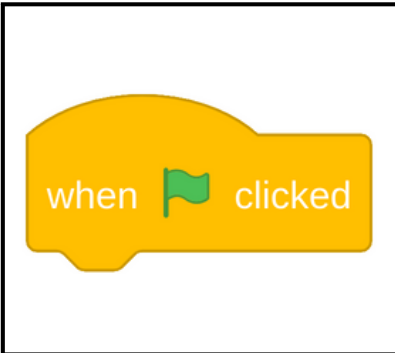
Highlight the strength of this activity, which lies in its ability to **make abstract concepts concrete: a loop becomes a repeated movement, a condition transforms into a choice of direction, a sequence manifests itself as a series of steps**.

This interdisciplinary approach, combining technology and physical expression, not only allows for better **memorization of concepts** but also contributes to the **well-being of learners by transforming learning into a fun and non-judgmental experience**.

Discuss the different dimensions of learning: **cognitive dimension** (understanding programming concepts), **physical dimension** (coordination and body expression), **creative dimension** (invention of new blocks) and **social dimension** (collaboration and communication).

Conclude by broadening the view of programming, presenting it not just as writing code for a computer, but as a **way of thinking, organizing ideas, solving problems, and creating**—a true language of expression that can take many forms, including movement.





REPEAT

**START WITH THE
GREEN FLAG**

WAIT

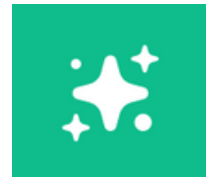
SOUND

CHANGE

**START WITH THE
SPACE KEY**

SPEECH BUBBLE

WALK



**DELETE A
SPRITE**

SURPRISE

**CHOOSING A
SPRITE**

**CHOOSE A
BACKGROUND**

CHOOSE A SOUND

DRAW

FILL

**RANDOM
POSITION**



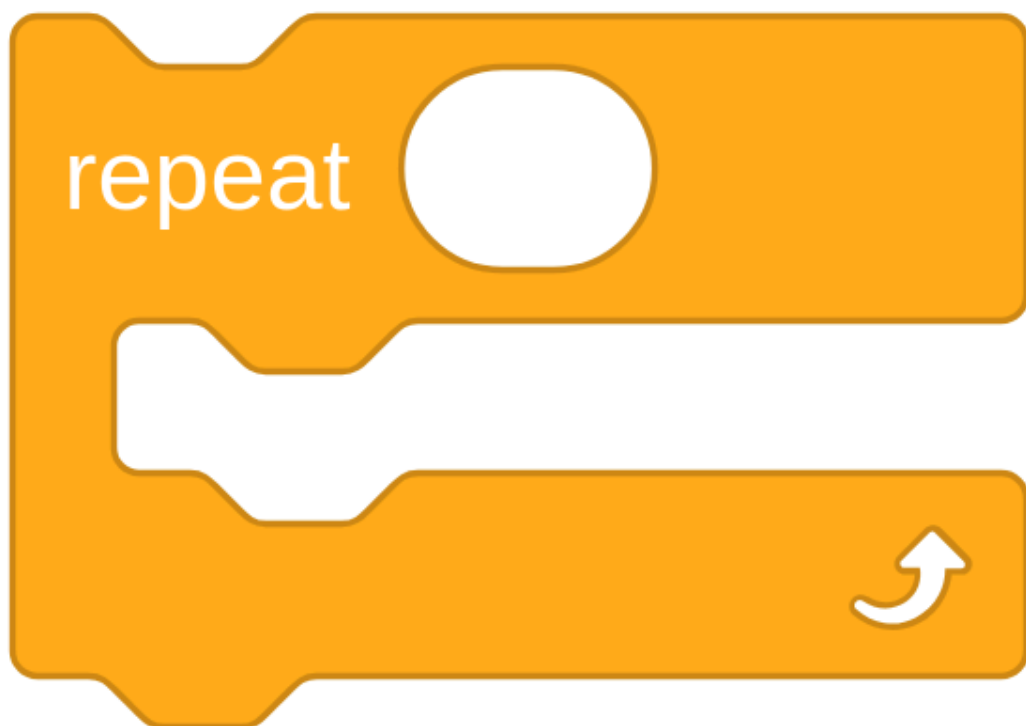
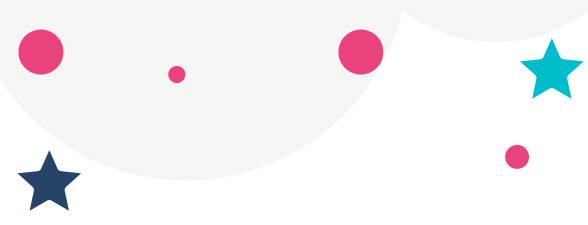
go to random position ▼

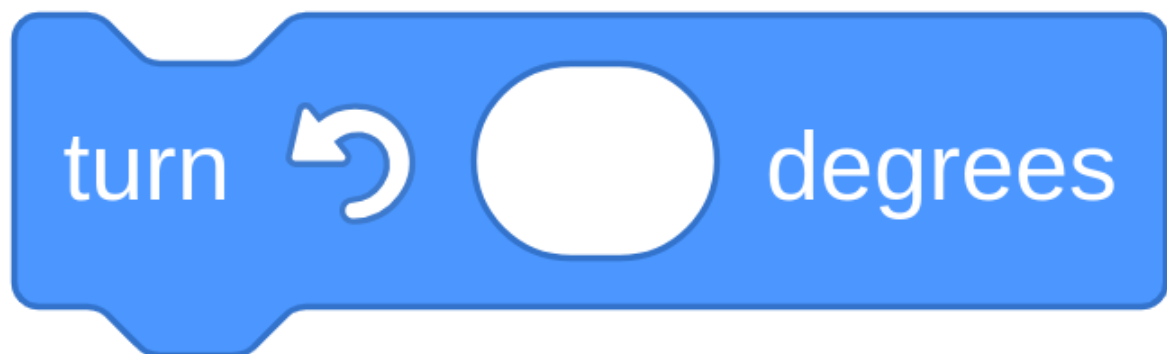
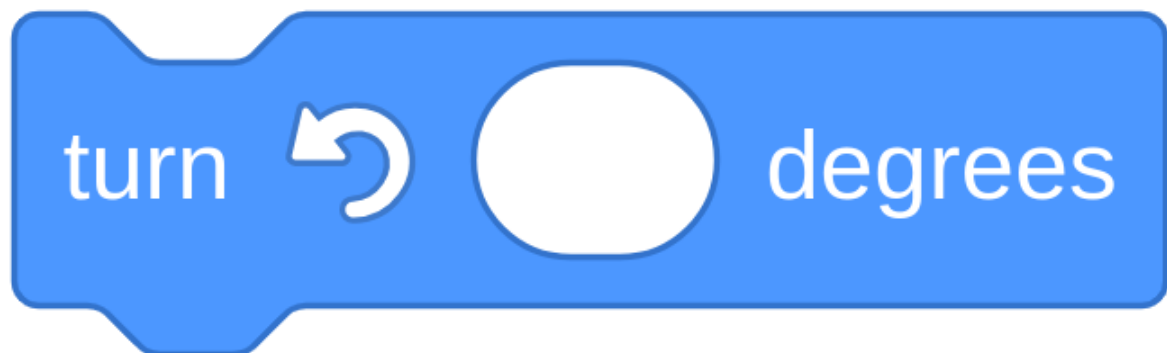
when  clicked

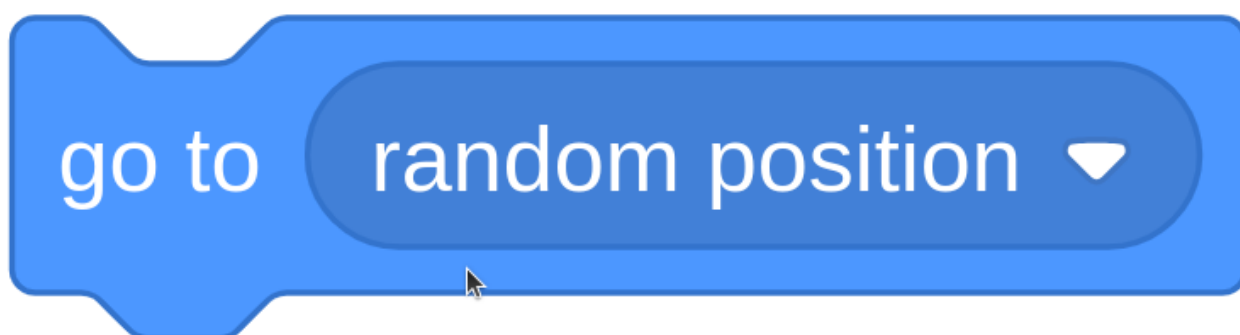
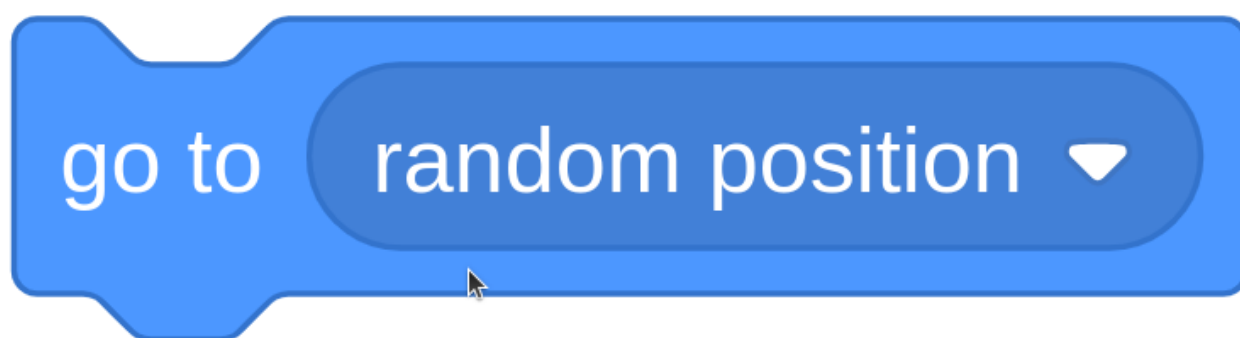
forever













stop all sounds

change volume by

start sound

Meow ▼

play sound

Meow ▼

until done



